E-Commerce Sales Data Analysis

Business Problem

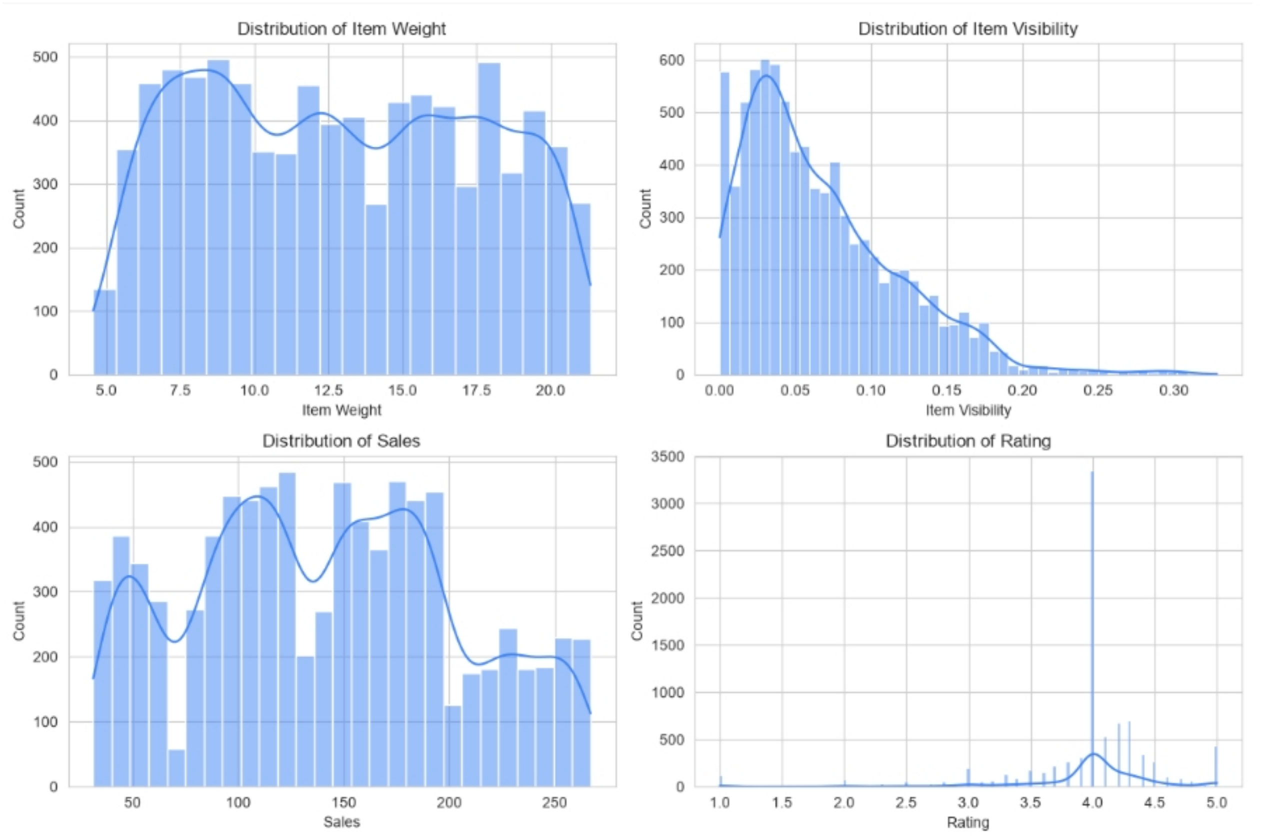
E-commerce store operates across multiple outlet types, sizes, and location tiers, but sales performance varies significantly between them, and it's unclear which factors — outlet format, item category, or product attributes — drive the biggest differences in revenue. This lack of visibility makes it difficult for business teams to prioritize inventory, allocate resources, and identify underperforming outlets or item categories. The goal of this analysis is to:

- Analyze sales data to uncover key performance drivers
- Build interactive dashboard for real-time sales insights
- Enable data-driven decisions across outlets and categories

Exploratory Data Analysis Insights

Summary Statistics

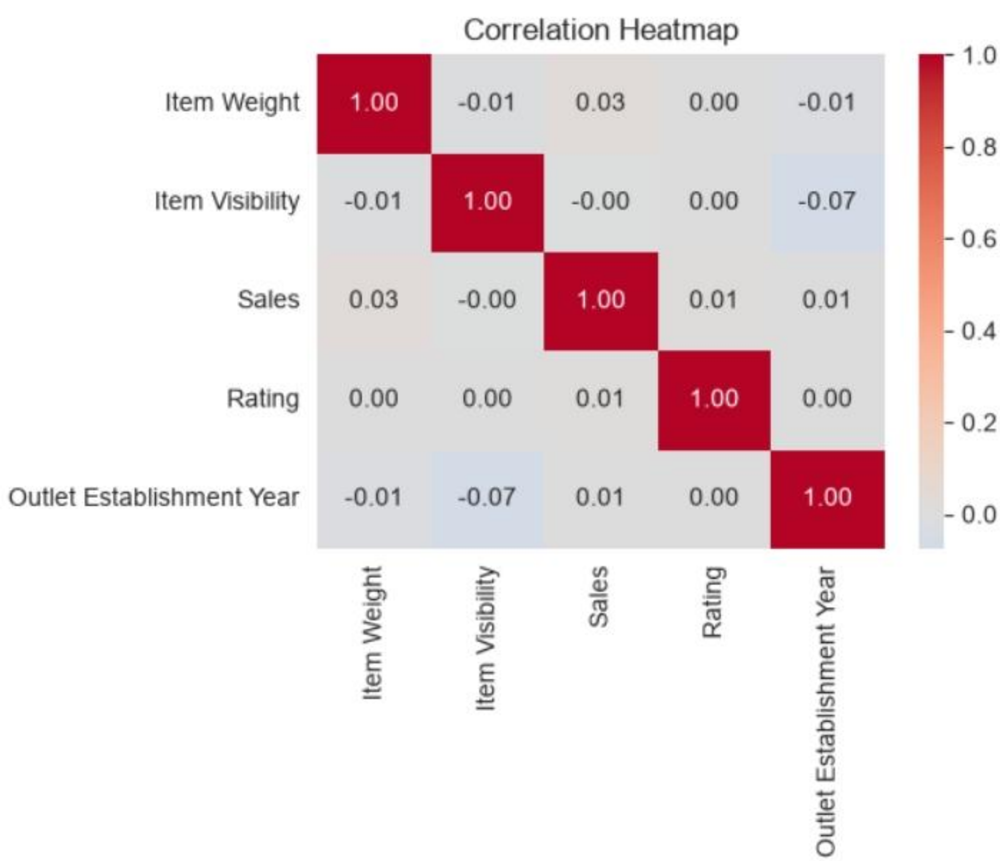
	Outlet Establishment Year	Item Visibility	Item Weight	Sales	Rating
count	8523.000000	8523.000000	7060.000000	8523.000000	8523.000000
mean	2010.831867	0.066132	12.857645	140.992782	3.965857
std	8.371760	0.051598	4.643456	62.275067	0.605651
min	1998.000000	0.000000	4.555000	31.290000	1.000000
25%	2000.000000	0.026989	8.773750	93.826500	4.000000
50%	2012.000000	0.053931	12.600000	143.012800	4.000000
75%	2017.000000	0.094585	16.850000	185.643700	4.200000
max	2022.000000	0.328391	21.350000	266.888400	5.000000



- Item Weight ranges from 4.56 to 21.35, averaging 12.86 units, with a fairly uniform spread across 5–20, showing no strong skew or dominant weight category.
- Item Visibility is heavily right-skewed, averaging just 0.066, with most products clustered between 0.00–0.10, meaning the majority of items have low shelf visibility overall.
- Sales range widely from ₹31.29 to ₹266.89, averaging ₹140.99, with a multimodal right-skewed distribution showing peaks around ₹100–130 and ₹150–200 ranges.
- Rating shows an extremely sharp spike near 4.0, with mean 3.97, suggesting most items share a common default rating rather than varied customer feedback.

- Outlet Establishment Year spans 1998 to 2022, averaging around 2011, indicating outlets have been operating across roughly two decades with fairly even distribution.
- The 25th percentile for Sales is ₹93.83, while the 75th percentile is ₹185.64, showing the middle 50% of sales fall within a ₹92 range.
- Item Weight has only 7,060 non-null values out of 8,523 total records, meaning about 1,463 entries were missing before median-based imputation was applied.
- Standard deviation for Sales is 62.28, indicating considerable variability in individual sale amounts across different items, outlets, and categories in the dataset.

Correlation Insights

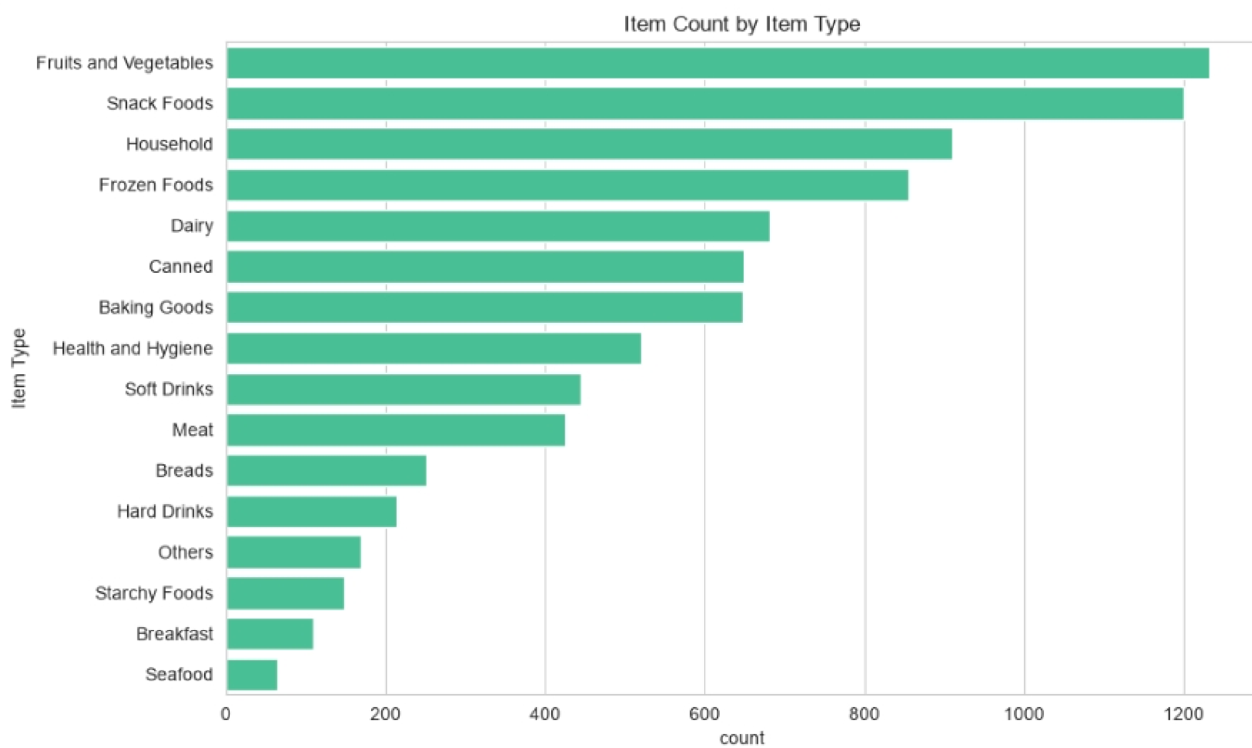


- **Item Weight vs. Sales:** Weak correlation (0.03), indicating that heavier or lighter items do not significantly influence total sales revenue generated.

- **Item Visibility vs. Outlet Establishment Year:** Weak negative correlation (-0.07), suggesting newer outlets tend to display items with marginally lower shelf visibility placement.
- **Item Visibility vs. Sales:** Near-zero correlation (-0.00), confirming that how visibly a product is placed does not meaningfully drive sales performance.
- **Rating vs. Sales:** Negligible correlation (0.01), showing that customer or item ratings have essentially no measurable impact on total sales figures.
- **Outlet Establishment Year vs. Sales:** Weak correlation (0.01), indicating outlet age has little to no effect on how much revenue an outlet generates.

Research Questions & Key Findings

1. Item Count by Item Type



- **Fruits and Vegetables** is the most stocked item type with roughly 1,230 items, followed closely by Snack Foods at around 1,200 items.

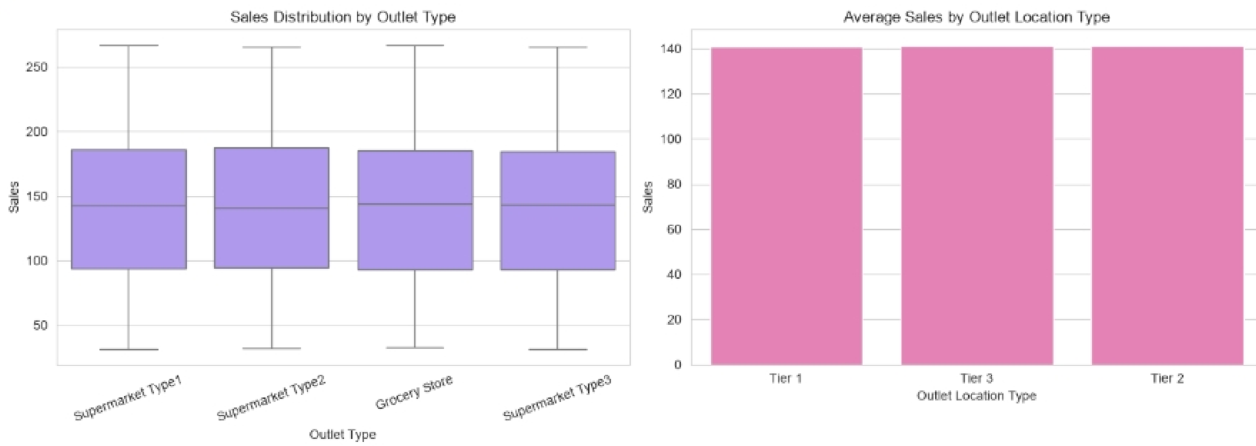
- **Household** and **Frozen Foods** are the next highest categories, with approximately 910 and 850 items respectively, showing strong mid-tier product representation.
- **Dairy, Canned, and Baking Goods** form a cluster of moderately stocked categories, each holding roughly 650–680 items in the inventory.
- **Health and Hygiene** and **Soft Drinks** have moderate counts near 520 and 440 items, positioned in the middle of the overall distribution.
- **Seafood** is the least stocked category with only about 65 items, followed by Breakfast and Starchy Foods, all under 200 items each.

2. Sales Analysis by Outlet Attributes

Outlet Type	mean	sum	count
Supermarket Type1	141.213894	787549.8868	5577
Grocery Store	140.294688	151939.1470	1083
Supermarket Type2	141.678634	131477.7724	928
Supermarket Type3	139.801791	130714.6746	935

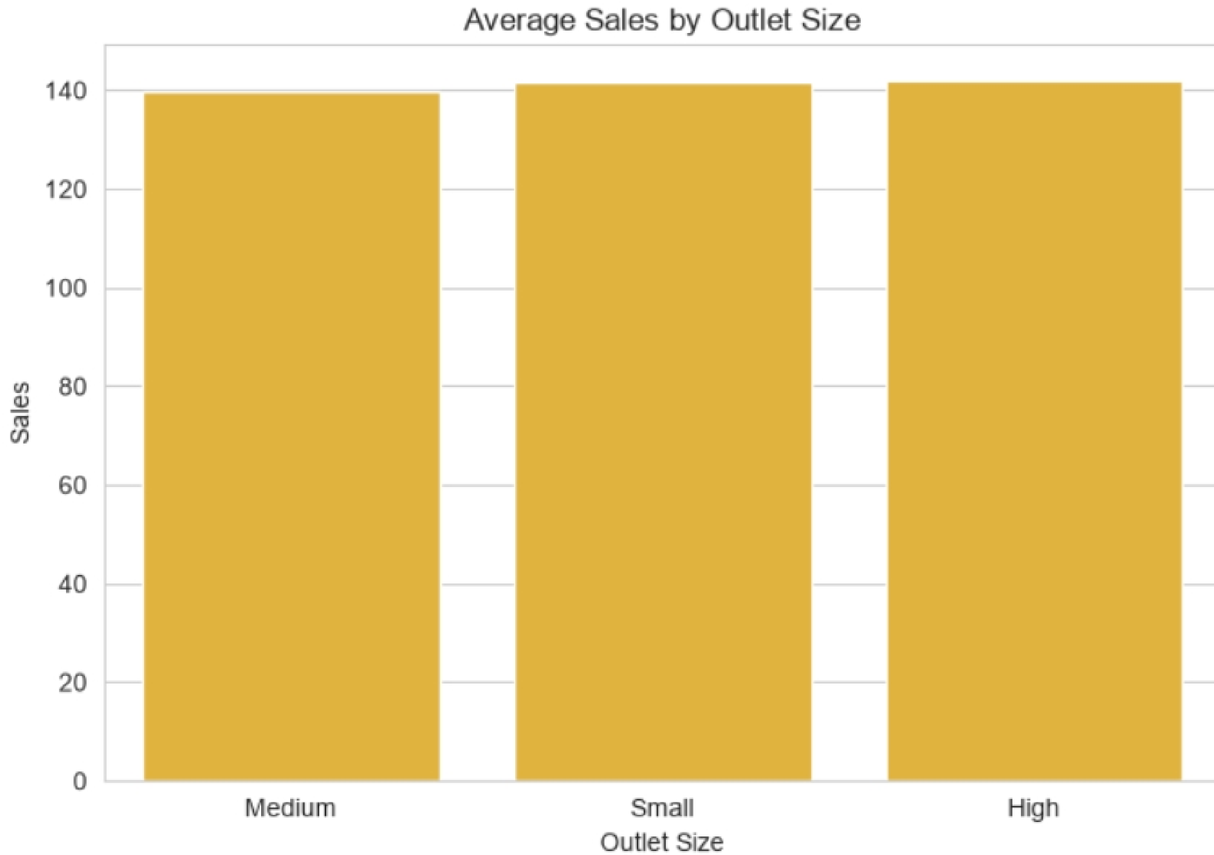
- **Sales distribution** is nearly identical across all four outlet types, with medians around ₹140 and similar interquartile ranges, showing consistent performance regardless of format.
- **Average sales by location tier** are remarkably uniform across Tier 1, Tier 2, and Tier 3, all hovering near ₹140, indicating tier has minimal impact.
- **Supermarket Type1** generates the highest total sales sum at ₹787,549, driven largely by its much larger transaction count of 5,577 records.

- **Supermarket Type2** has the highest average sales per item at ₹141.68, marginally outperforming other outlet types despite having the fewest transactions (928).
- **Grocery Store** and **Supermarket Type3** show similar transaction counts (1,083 and 935) and comparable average sales (~₹140), reflecting balanced performance across smaller formats.



3. Avg Sales by Outlet Size

- **High** outlet size shows the highest average sales at approximately ₹142, marginally outperforming Small and Medium outlet size categories overall.
- **Small** outlets closely follow with average sales near ₹141.5, showing almost identical performance to High-sized outlets despite differing scale.
- **Medium** outlets have the lowest average sales at roughly ₹139.5, though the difference from other sizes remains minimal and insignificant.
- All three outlet sizes cluster tightly between ₹139–142, indicating outlet size has **negligible impact** on average sales performance overall.



4. Avg Sales by Item Type

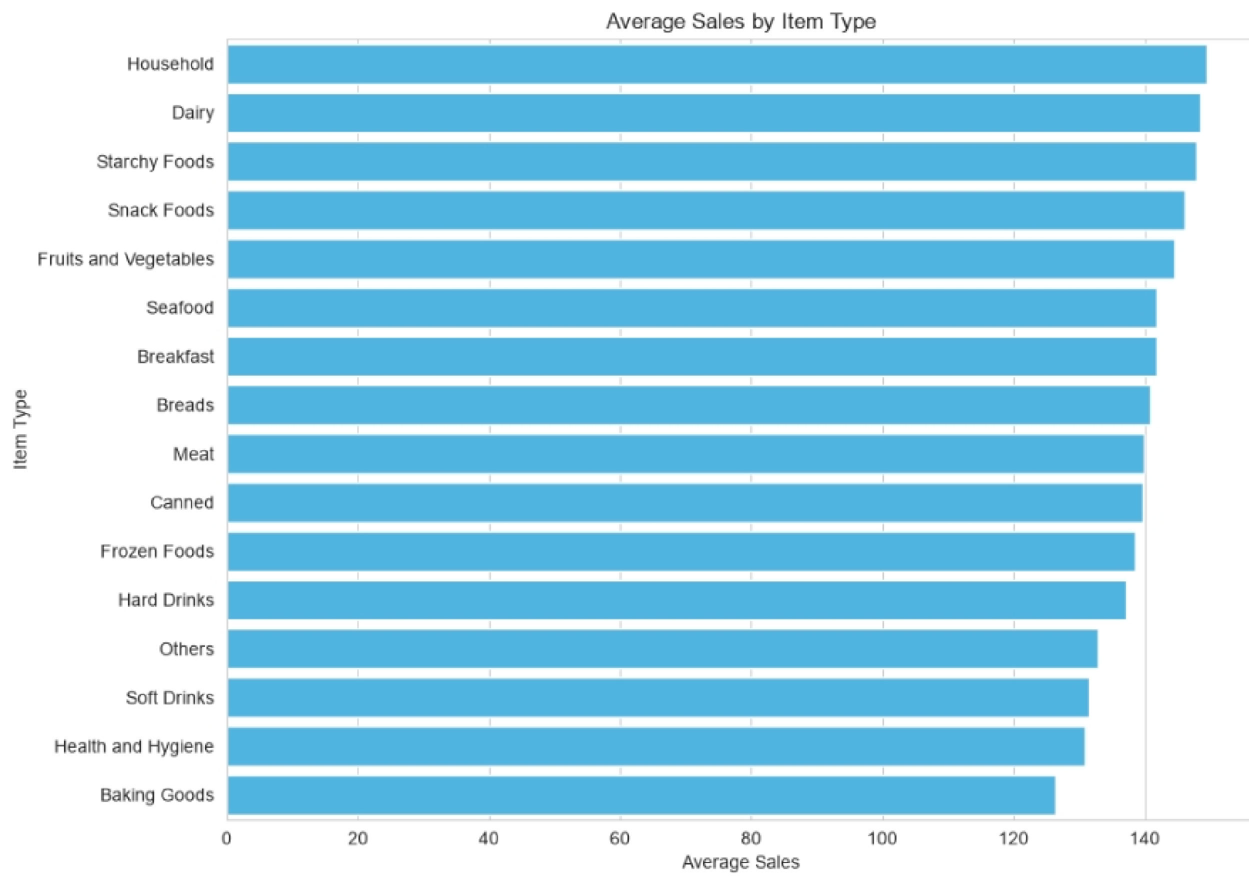
Household items generate the highest average sales at approximately ₹150, narrowly leading Dairy and Starchy Foods among all item categories.

Dairy, Starchy Foods, and Snack Foods form a top-tier cluster, each averaging around ₹146–149, showing consistently strong sales performance overall.

Fruits and Vegetables, Seafood, and Breakfast hold mid-range positions with average sales near ₹142–145, despite differing stock volumes significantly.

Baking Goods has the lowest average sales at roughly ₹127, trailing behind Health and Hygiene and Soft Drinks in overall performance.

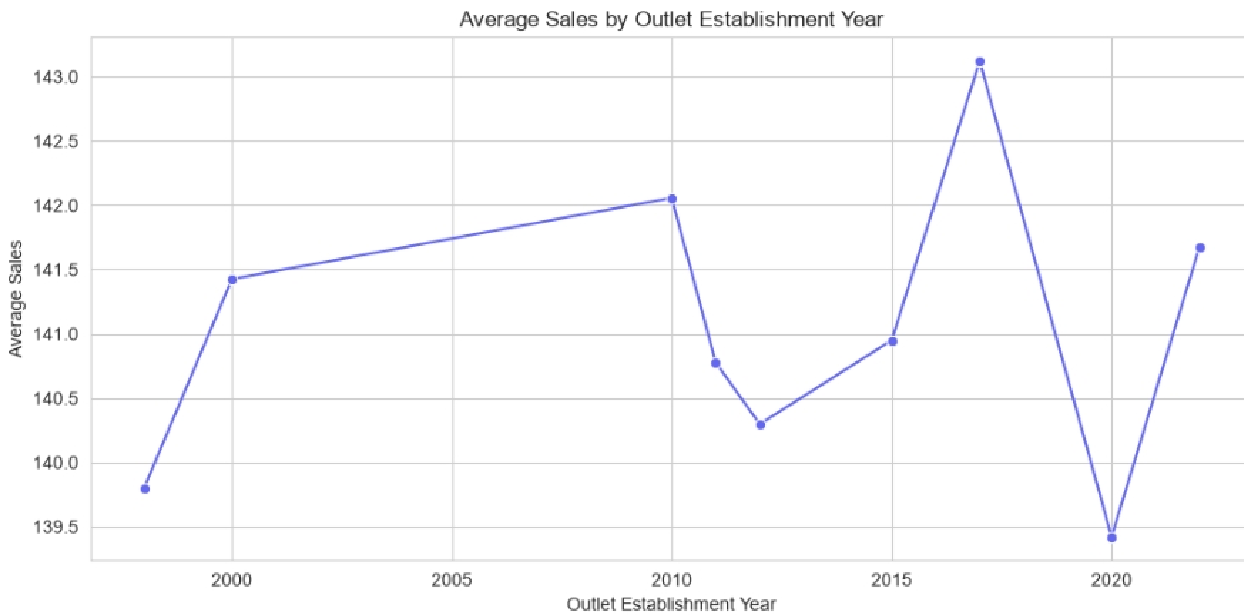
Sales range is fairly narrow (₹127–150) across all item types, suggesting **item category alone** has limited influence on average sales value.



5. Avg Sales by Outlet Establishment Year

- **2017** shows the highest average sales at approximately ₹143.1, marking a sharp peak compared to surrounding establishment years in the dataset.
- **2020** records the lowest average sales at roughly ₹139.4, representing a significant dip right before recovering again by 2022.
- Sales rose steadily from **1998 to 2010**, increasing from ₹139.8 to ₹142.1, suggesting older outlets gradually built stronger sales performance.

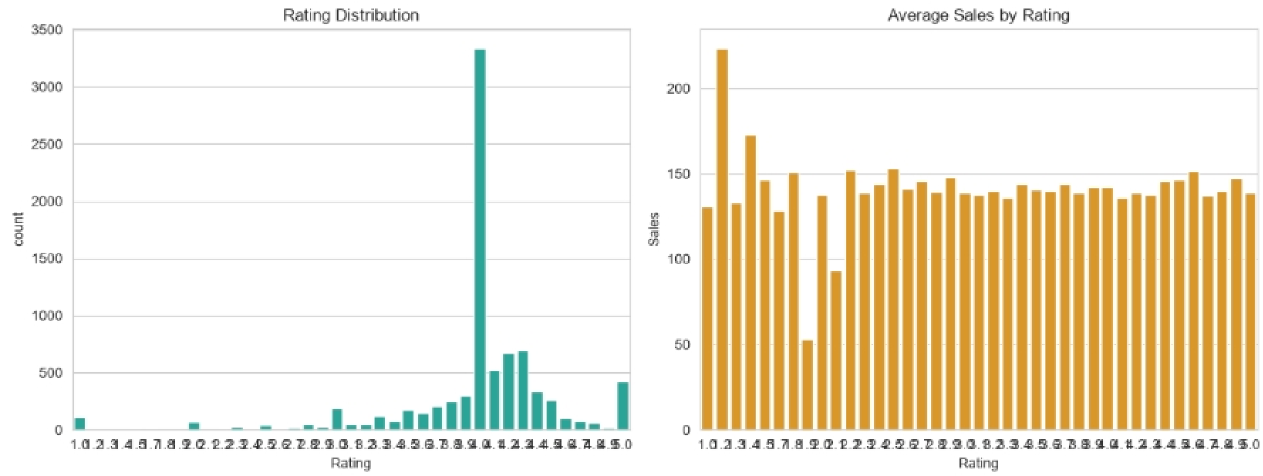
- A noticeable **decline occurred between 2010 and 2012**, dropping from ₹142.1 to ₹140.3, before slowly climbing back up through 2015.
- Overall, average sales fluctuate narrowly between **₹139.4 and ₹143.1**, indicating establishment year has only a mild, non-linear effect on sales.



6. Rating Analysis

- Rating of 4.0 dominates the distribution with over 3,300 occurrences, far exceeding all other rating values combined in the dataset.
- Most other ratings show relatively low counts, mostly under 500, with a smaller secondary spike near 5.0 at around 400 occurrences.
- Average sales by rating fluctuate significantly, ranging from as low as ₹50 to peaks above ₹225, showing no clear linear trend.
- Rating around 1.3 shows the highest average sales near ₹225, though this likely reflects a small sample size rather than a true pattern.

- Most rating values between 2.0 and 5.0 show average sales clustering around ₹130–150, indicating rating has limited real influence on sales.



Overall Analysis Summary

Key Numeric Patterns
Sales range from ₹31 to ₹267 (avg ₹141) with a right-skewed, multimodal distribution. Item Visibility is heavily skewed toward low values (avg 0.066), while Rating is oddly concentrated at 4.0 — suggesting it may be a default/assigned value rather than genuine customer feedback.

Correlation Findings
None of the numeric features (Weight, Visibility, Rating, Establishment Year) show meaningful correlation with Sales (all |r| < 0.03). This means **sales aren't explained by simple numeric relationships** — the real drivers lie in categorical/structural factors.

Category-Level Drivers

- **Outlet Type, Size, and Location Tier** all show remarkably flat average sales (~₹139–142) — none of these outlet-level attributes meaningfully separate high vs. low performers.
- **Item Type** shows more variation: Household, Dairy, and Starchy Foods lead (~₹146–150), while Baking Goods trails (~₹127) — a ~₹23 spread, the widest signal found in the dataset.
- **Establishment Year** shows mild non-linear fluctuation (₹139–143) with a 2017 peak and 2020 dip, but no strong trend.

Suggested Next Steps

1. **Investigate the Rating anomaly** — the spike at 4.0 needs a root-cause check (data entry default? survey design?) before treating Rating as a usable feature.
2. **Run statistical significance tests** (ANOVA / Kruskal-Wallis) on Item Type vs. Sales to confirm whether that ₹23 spread is statistically meaningful or just noise.
3. **Engineer new features** — e.g. Sales per Visibility, Weight-to-Sales ratio, or Outlet Age (2026 − Establishment Year) — since raw numeric fields show no correlation.
4. **Segment by combinations**, not single variables — e.g. Item Type × Outlet Type, since individual factors are weak but interactions may reveal stronger patterns.